INMO(4^{th}) – 1989

Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
- Ruler and compass allowed.
- Answer all the questions.

1. Prove that the polynomial $f(x) = x^4 + 26x^3 + 56x^2 + 78x + 1989$ can be expressed as product $f(x) = p(x)q(x)$, where $p(x), q(x)$ are polynomials with integral coefficients.
2. Let a, b, c, d are any four real numbers not equal to zero. Prove that the roots of the polynomial $f(x) = x^6 + ax^3 + bx^2 + cx + d$ cannot be real.
3. Let A denote a subset of the set {1, 11, 21, 31, ….., 541, 551} having the property that no two elements of A add up to 552. Prove that A cannot have more than 28 elements.
4. Determine with proof, all the positive integers n for which:
   (a) n is not a square of any integer and
   (b) $[\sqrt{n}]$ divides n^2 (where [x] greatest integer function).
5. For positive integers n, define A(n) to be $\frac{(2n)!}{(n!)^2}$. Determine the sets of positive integers n for which:
   (a) A(n) is an even number.
   (b) A(n) is a multiple of 4.
6. Triangle ABC has interior I and the incircle touches BC, CA at D, E respectively. Let BI meet DE at G. Show that AG is perpendicular to BC.
7. Let A be one of the two points of intersection of two circles with centre X and Y respectively. The tangents at A to the two circles meet the circles again at B, C. Let a point P be located so the PXAY is a parallelogram. Show that P is also circumcentre of triangle ABC.

******
****
**